

1.3 Maximum Angular Displacement And Angular Velocity

The torsional motion of the driven gear is governed by
 $J_0 \ddot{\theta} + k_t \theta = M_t(t)$

Given Parameter :

- $J_0 = 0,10 \text{ kg.m}^2$
- $G = 79,3 \text{ GPa} = 79,3 \times 10^9 \text{ Pa}$
- $d = 0,050 \text{ m}$
- $L = 1,00 \text{ m}$
- $M = 1000 \text{ N.m}$

The polar second moment of area of the solid shaft

$$J_p = \frac{\pi d^4}{32}$$

$$J_p = \frac{\pi (0,05)^4}{32}$$

$$J_p = 6,136 \times 10^{-7} \text{ m}^4$$

The torsional Stiffness

$$k_t = \frac{G J_p}{L}$$

$$k_t = \frac{(79,3 \times 10^9)(6,136 \times 10^{-7})}{1}$$

$$k_t = 48\,657,87 \text{ N.m/rad}$$

The initial steady-state angular displacement

$$\theta_0 = \frac{M}{k_t} = \frac{1000}{48657,87}$$

$$\theta_0 = 0,02055 \text{ rad}$$

Using the broken-tooth torque disturbance in the Simulink model, the simulated response gives

Maximum angular displacement :
 $\theta_{max} \approx 0,0233 \text{ rad}$

Maximum angular velocity :
 $|\dot{\theta}|_{max} = 1,92 \text{ rad/s}$

1.4 Dominant Torsional Vibration Frequency and Resonance

The natural angular frequency of the torsional system

$$\omega_n = \sqrt{\frac{K_t}{J_o}}$$
$$\omega_n = \sqrt{\frac{48657,87}{0,10}}$$

$$\omega_n = 697,55 \text{ rad/s}$$

Convert natural frequency to Hz

$$f_n = \frac{\omega_n}{2\pi} = \frac{697,55}{2\pi}$$

$$f_n = 111,02 \text{ Hz}$$

Rotational Frequency

$$f_r = \frac{1000}{60}$$

$$f_r = 16,67 \text{ Hz}$$

The 7th harmonic

$$f_7 = 7(16,67)$$

$$f_7 = 116,67 \text{ Hz}$$

$$\Delta f = 116,67 - 111,02$$

$$\Delta f = 5,65 \text{ Hz}$$

